

Wireshark Network Traffic Analysis Report

This project demonstrates the use of Wireshark to capture and analyze network traffic. The objective is to observe data packets, apply filters, examine packet details, and identify potential security concerns.

Step 1: Set Up Wireshark

Wireshark was downloaded from the official website and installed. The network interface was selected to begin monitoring.

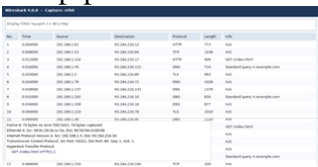
Step 2: Capture Network Traffic

Traffic was captured by starting Wireshark and performing activities such as browsing a website, sending an email, and accessing a shared resource. The capture was stopped after sufficient data was collected.

Step 3: Filter Captured Traffic

Filters were applied to narrow the data:

- http → Display HTTP traffic
- ip.addr == 192.168.1.1 → Show packets involving a specific IP address
- tcp.port == 80 → Focus on HTTP traffic

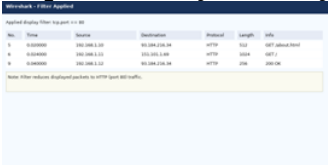


No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
2	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
3	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
4	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
5	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
6	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
7	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
8	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
9	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
10	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
11	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
12	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
13	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
14	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
15	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
16	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /
17	0.00000	192.168.1.10	192.168.1.1	HTTP	121	GET /

Step 4: Analyze Packet Details

Individual packets were analyzed to extract source and destination IP addresses, protocol details (HTTP requests, DNS queries), and anomalies such as suspicious IPs or repeated failed connections.

Step 5: Identify Security Concerns: Potential security issues identified included:



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- Unencrypted HTTP traffic observed
- Communication with unexpected IP addresses
- High volumes of repetitive traffic suggesting possible scanning activity

Packet Details - Frame 6

Hypertext Transfer Protocol
GET /index.html HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
Accept: text/html,application/xhtml+xml

Transmission Control Protocol
Source Port: 54321
Destination Port: 80
Sequence Number: 1
Acknowledgment Number: 1

Summary & Recommendations

Wireshark effectively revealed the structure and flow of network communications. To enhance security, it is recommended to enforce HTTPS, monitor unusual IP communications, and set alerts for traffic anomalies.